THE S-MORPHISM FUNCTOR FROM A FINITE LOCALLY FREE SOURCE INTO AN AFFINE TARGET IS A SCHEME AFFINE OVER THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 20170-20176 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 20178-20190, the complete author proof (11 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Extends the lane's finite-free Hom-scheme card (already collected) from free to locally free modules by descent-style gluing.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Chapter lemma hom-from-finite-free-into-affine (the finite free algebra case: Mor_S represented by Spec of a quotient of the symmetric algebra), Schemes, lemma glue-functors (the subfunctor gluing criterion), and basic support arguments for functors on Sch/S.

Source reference. Original lemma 05Y6: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 20100–20105. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01JJ: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 20190–20190. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $Z \to S$ and $X \to S$ be morphisms of schemes. If $Z \to S$ is finite locally free and $X \to S$ is affine, then $Mor_S(Z, X)$ is representable by a scheme affine over S.*

Proof. Choose an affine open covering $S = \bigcup U_i$ such that $\Gamma(Z \times_S U_i, \mathcal{O}_{Z \times_S U_i})$ is finite free over $\mathcal{O}_S(U_i)$. Let $F_i \subset Mor_S(Z, X)$ be the subfunctor which assigns to T/S the empty set if $T \to S$ does not factor through U_i and $Mor_S(Z, X)(T)$ otherwise. Then the collection of these subfunctors satisfy the conditions (2)(a), (2)(b), (2)(c) of Schemes, Lemma 01JJ which proves the lemma. Condition (2)(a) follows from Lemma 05Y6 and the other two follow from straightforward arguments. □

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.